RECAP: Reaction Completion and Prediction for Biochemical Reactions with Incomplete Inputs

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Abstract

Biochemical reaction prediction underpins enzyme annotation, pathway reconstruction, and the design of non-native biosynthetic routes. Reaction language models pretrained on organic chemistry, however, transfer poorly to metabolic reactions, whose transformations involve cofactors, charged species, variable protonation states, and multi-product stoichiometry. In this work we evaluate three strategies for adapting ReactionT5 to metabolic reaction prediction using MetaNetX dataset (approximately 19,000 valid entries): (i) supervised fine-tuning, (ii) functional-group knowledge-graph–guided tuning, and (iii) LongT5-based long-context modelling. Supervised fine-tuning yielded the largest improvement in top-1 product recovery, raising all-product accuracy from 0.003 to 0.173 and any-product accuracy from 0.025 to 0.586. Neither the knowledge-graph augmentation nor the LongT5 architecture improved top-1 accuracy beyond standard supervised fine-tuning. Although the adapted models generated products with higher structural similarity to the references, stoichiometric balance remained low as the best balance score reached only 0.125. Element-wise analysis revealed that conservation improved most for N, P, and S, while H, O, and C remained the principal sources of imbalance. These results demonstrate that supervised domain adaptation meaningfully improves metabolic product prediction, yet implicit learning from curated reactions alone is insufficient to guarantee elemental and stoichiometric correctness due to small tuning dataset. Therefore, future models will use a larger dataset to fine-tune the language model.

Keywords

biochemical reaction completion; reaction prediction; MetaNetX; SMILES; stoichiometric balance; chemical consistency; constrained generation; ReactionT5; knowledge graph; pathway design, language model

1 Introduction

Predicting biochemical reactions of the molecular transformations that connect metabolites through enzyme-catalyzed steps is a foundational problem at the intersection of computational chemistry, metabolic engineering, and pathway design. Whether the goal is to annotate enzyme function, reconstruct metabolic pathways, or engineer non-native biosynthetic routes in microbial hosts, the ability to accurately forecast reaction outcomes remains essential [1–3].

Two broad families of methods currently dominate such effort. Rule-based approaches apply transformation patterns, bond-change operators, or curated reaction rules drawn from known biochemistry. Frameworks such as BNICE, RetroPath2.0, RetroRules, and the novoStoic pathway-design suite exemplify this direction [2–7]. Their principal strength is chemical interpretability where every prediction traces back to an explicit, human-readable rule. However, their predictive scope is inherently bounded by the coverage and diversity of the underlying rule library. Missing reaction classes, incomplete cofactor logic, and overly generic transformation templates can exclude biologically feasible routes that fall outside the library's reach.

Neural sequence-based approaches sidestep many of these limitations by casting reaction prediction as a translation problem between molecular strings, most commonly SMILES representations [8]. Models such as Molecular Transformer, Chemformer, T5Chem, ReactionT5, and RSGPT learn reactant–product mappings directly from large reaction corpora, enabling generalization well beyond predefined templates [9–13]. However, because these models generate output tokens sequentially, they can produce syn-

tactically valid SMILES that nonetheless violate elemental balance, stoichiometric consistency, or redox logic at the full-reaction level. Advances in constrained decoding, atom-mapping-aware training, post-hoc balancing, and reaction-specific pretraining have improved chemical validity, but chemical validity alone is an insufficient guarantee of biologically meaningful predictions.

A deeper challenge arises from the fundamental mismatch between the chemistry these models learn and the chemistry they are asked to predict. Most pretrained reaction language models are trained on organic-chemistry or patent-derived datasets dominated by laboratory reagents, protecting-group manipulations, and small-molecule synthesis [9, 10, 12, 14]. Metabolism, by contrast, operates under a distinct molecular logic as the recurring cofactors, enzyme-mediated group transfers, tightly coupled redox and energy balances, pathway-level context, and organism-specific constraints all shape the space of feasible transformations [1–3]. Consequently, strong performance on organic-chemistry benchmarks does not reliably transfer to metabolic reaction prediction in organisms such as *Escherichia coli* or *Pseudomonas putida*.

Cofactors is the primary domain gap for the reaction predictions. Molecules such as NADH, NADPH, coenzyme A, ATP, S-adenosylmethionine, flavin adenine dinucleotide, and pyridoxal phosphate are far more than passive labels attached to reaction entries. They actively mediate hydride transfer, electron shuttling, acyl activation, phosphoryl-group transfer, and the stabilization of reactive intermediates. Any model that aspires to accurate metabolic prediction must therefore capture these cofactor-dependent chemical contexts, not merely the structures of the principal substrates and products. SMILES-based language models fails constantly as cofactor SMILES strings are long, highly recurrent, and structurally complex, often dwarfing the strings of the central metabolites they accompany. The resulting increase in sequence length complicates tokenization and makes it harder for learned embeddings to preserve the chemically decisive substructures within a cofactor. This representation bottleneck becomes acute when the model must distinguish closely related transformations. For example, NADH-dependent versus NADPH-dependent reductions, ATP-driven activation steps, or coenzyme A-mediated acyl transfers.

In this work, we systematically evaluate three strategies for bridging this gap: supervised fine-tuning on a MetaNetX-derived biochemical dataset, functional-group knowledge-graph-guided tuning, and LongT5-based long-context modeling. Supervised fine-tuning alone yields substantial improvements in top-1 product recovery over the pretrained baseline, demonstrating that reaction language models trained on organic chemistry can be efficiently adapted to metabolic prediction with a comparatively modest domain-specific dataset. By contrast, the knowledge-graph augmentation and LongT5 architecture do not deliver additional gains beyond standard fine-tuning under the evaluation conditions examined here. Importantly, improved product recovery does not guarantee chemically complete reactions where the elemental and stoichiometric balance remain limited in the absence of explicit balancing constraints. These findings pinpoint cofactor representation, reaction balance enforcement, and biochemical constraint integration as the principal bottlenecks in transferring language model logic to enzymatic reaction predictions. Achieving reliable biochemical reaction prediction will therefore demand not only domain-specific fine-tuning but also learning objectives or post-generation procedures that may enforce stoichiometry, cofactor identity, redox consistency, and enzyme-relevant biochemical constraint. However, the data itself is incomplete, and strict enforcement could therefore lead to erroneous training signals.

2 Method

Three strategies were evaluated for adapting the model to metabolic reaction prediction: supervised fine-tuning, functional group knowledge graph-guided tuning, and LongT5-based long-context modeling. These approaches were designed to improve biochemical plausibility and sequence stability, but they did not explicitly enforce elemental or stoichiometric balance in the training objective. Strict hard enforcement was avoided at this stage because metabolic reactions often include cofactors, protonation states, charged species, and database-dependent stoichiometric conventions, making rigid balance constraints sensitive to annotation noise and incomplete reaction specifications [15]. Instead, balance was treated as an implicit data-level constraint

through curated reactions and chemically informed adaptation, while explicit balance correction can be applied later through post-generation filtering, atom mapping, or a balance-aware objective [2].

2.1 Supervised fine-tuning of ReactionT5

ReactionT5 [12], a language model originally developed for organic reaction prediction, was adapted for metabolic and enzymatic reaction prediction through supervised fine-tuning. The supervised fine-tuning stage used the pretrained ReactionT5v2-forward model as the initialization. The training and validation reaction datasets were first filtered to remove entries without valid reactant or product fields and were then standardized using the same preprocessing pipeline used for ReactionT5 training. Each reaction was formulated as a conditional sequence generation task, in which the model input was constructed by concatenating the reactant and reagent fields:

$$x_i = \texttt{REACTANT:}R_i\texttt{REAGENT:}G_i,$$

where R_i denotes the reactant string and G_i denotes the reagent string. The corresponding target sequence was the product string:

$$y_i = P_i.$$

The fine-tuned sequence-to-sequence model defines the conditional probability of the product sequence as

$$p_\theta(y_i \mid x_i) = \prod_{t=1}^{T_i} p_\theta\left(y_{i,t} \mid y_{i,<t}, x_i\right),$$

where $y_{i,t}$ is the product token at decoding step t, $y_{i,<t}$ denotes the preceding product tokens, T_i is the target sequence length, and θ represents the model parameters. The supervised fine-tuning objective was the token-level negative log-likelihood:

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\sum_{i=1}^{N}\sum_{t=1}^{T_i} \log p_\theta\left(y_{i,t} \mid y_{i,<t}, x_i\right).$$

Training was performed using a sequence-to-sequence trainer with dynamic padding and generation-based validation. During evaluation, beam search was used to generate product predictions, and model performance was measured using prediction accuracy. The best checkpoint was selected according to validation accuracy, and early stopping was applied to limit overfitting. When additional chemically similar reactions were provided, they were appended to the training set before fine-tuning to increase the representation of related transformations.

Elemental and stoichiometric balance were not explicitly enforced in the supervised fine-tuning objective. Instead, these constraints were treated as data-level properties of the curated reaction dataset. Therefore, the model learned balance-related patterns only implicitly from the training examples. Because the optimization objective maximizes token-level likelihood, it does not guarantee that generated products conserve atoms, charge, or stoichiometric coefficients.

2.2 SFT with functional group knowledge graph-based tuning

To introduce chemically interpretable transformation priors, we constructed a reaction-centric functional group knowledge graph (FGKG) [16] in addition to the SFT fine-tuning. The graph represents reactions at the level of functional group changes rather than raw SMILES strings, providing an intermediate abstraction for common enzymatic transformations such as oxidation, hydrolysis, condensation, and amination.

Functional groups were assigned using a rule-based molecular graph annotation procedure adapted from FARM. Each molecule was parsed into an RDKit [17] graph, and atoms were labeled using local chemical rules based on atom type, neighboring atoms, bond order, formal charge, hydrogen count, ring membership, and fused-ring structure. Electronegativity was included as an atom-level chemical descriptor. For atom a, the electronegativity was defined as

$$\chi(a) = \chi_{Z(a)},$$

where $Z(a)$ is the element type, such as Carbon, Oxygen, and Nitrogen. For a bond between atoms a and b, the electronegativity difference was

$$\Delta\chi_{ab} = |\chi(a) - \chi(b)|.$$

This descriptor helps distinguish polar bonds and heteroatom-containing functional environments, including carbonyls, alcohols, amines, halides, sulfides, and phosphate-containing groups.

For each reaction i, functional group count vectors were computed for the substrate and

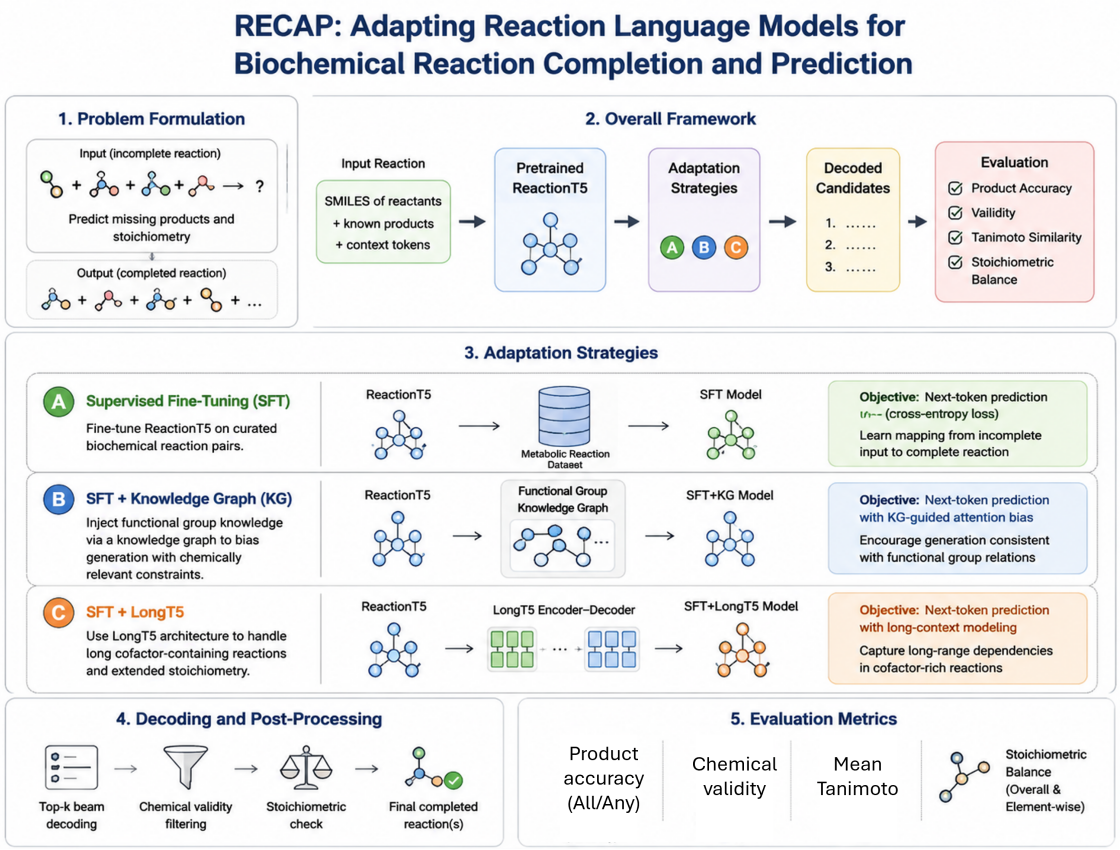


Figure 1: Graphical overview of the RECAP framework for biochemical reaction completion and prediction. The workflow starts from incomplete reaction inputs, adapts ReactionT5 using supervised fine-tuning, functional group knowledge graph-guided tuning, and LongT5-based long-context modeling, and evaluates generated reactions using product recovery, chemical validity, structural similarity, and stoichiometric balance.

product sides:

$$\mathbf{f}_i^{\text{sub}} = \sum_{m \in \mathcal{S}_i} \mathbf{f}(m), \qquad \mathbf{f}_i^{\text{prod}} = \sum_{m \in \mathcal{P}_i} \mathbf{f}(m),$$

where $\mathcal{S}_i$ and $\mathcal{P}_i$ are the substrate and product molecule sets, respectively. The net functional group change was then

$$\Delta \mathbf{f}_i = \mathbf{f}_i^{\text{prod}} - \mathbf{f}_i^{\text{sub}}.$$

Functional groups with negative entries in $\Delta \mathbf{f}_i$ were treated as consumed groups, whereas those with positive entries were treated as formed groups.

The FGKG was defined as

$$\mathcal{G}_{\text{FG}} = (\mathcal{V}_{\text{FG}}, \mathcal{E}_{\text{FG}}),$$

where $\mathcal{V}_{\text{FG}}$ is the set of functional group types and $\mathcal{E}_{\text{FG}}$ is the set of reaction-derived edges. Three relation classes were used: transformation, co-consumption, and co-formation. A transformation edge was added from each consumed group to each formed group in the same reaction:

$$(f_a, \text{trans}, f_b), \qquad f_a \in \mathcal{F}_i^{\text{cons}},\ f_b \in \mathcal{F}_i^{\text{form}}.$$

Co-consumption and co-formation edges were added between pairs of functional groups consumed or formed together:

$$(f_a, \text{coC}, f_b), \qquad f_a, f_b \in \mathcal{F}_i^{\text{cons}},$$

$$(f_a, \text{coF}, f_b), \qquad f_a, f_b \in \mathcal{F}_i^{\text{form}}.$$

The co-consumption and co-formation relations were treated as symmetric, and edge counts were accumulated across all reactions. Low-frequency edges were removed using a minimum-count threshold to reduce noise.

The resulting graph was represented as triples

$$\mathcal{T}_{\text{FG}} = \{(h, r, t) : h, t \in \mathcal{V}_{\text{FG}},\ r \in \mathcal{R}_{\text{FG}}\},$$

where $\mathcal{R}_{\text{FG}}$ contains transformation, co-consumption, and co-formation relations. Therefore, the FGKG converts reaction-level functional group changes into a compact multi-relational graph.

Following the FARM knowledge graph strategy, functional group embeddings can be learned from

these triples using a multi-relational embedding model. In a ComplEx formulation, each head, relation, and tail is represented by a complex vector, and the triple score is

$$s(h, r, t) = \mathrm{Re}\left(\mathbf{h}^\top \mathrm{diag}(\mathbf{r})\bar{\mathbf{t}}\right).$$

Positive triples were obtained from observed FGKG edges, and negative triples were generated by corrupting the head or tail entity. The graph embedding loss was

$$\mathcal{L}_{\mathrm{KG}} = \sum_{\mathcal{T}^+} \sum_{\mathcal{T}^-} \max\left[0, \gamma + s(h', r, t') - s(h, r, t)\right],$$

where γ is the margin, which is the required minimum gap between the score of a real graph relation and the score of a corrupted, false relation. This objective assigns higher scores to observed functional group relations than to corrupted relations.

The FGKG was used as a functional group-level plausibility prior rather than a strict balance constraint. It evaluates whether predicted functional group changes match reaction patterns observed in the training data, such as

$$\mathrm{alcohol} \rightarrow \mathrm{carbonyl}$$

and

$$\mathrm{ester} \rightarrow \mathrm{carboxyl} + \mathrm{hydroxyl}.$$

However, functional group consistency implies but does not guarantee conservation of atoms, charge, or stoichiometric coefficients. For example,

$$\mathrm{CH_3CH_2OH} \rightarrow \mathrm{CH_3CHO}$$

is a plausible alcohol oxidation pattern, but it is not balanced as written because two hydrogen atoms are missing. A balanced biochemical form requires an electron acceptor:

$$\mathrm{CH_3CH_2OH + NAD^+} \rightarrow \mathrm{CH_3CHO + NADH + H^+}.$$

Therefore, the FGKG improves reaction-type plausibility.

2.3 SFT with LongT5 adaptation for cofactor-containing reactions

Enzymatic reaction strings are longer than standard organic reaction strings as they include large cofactors and carrier molecules, such as $\mathrm{NAD^+/NADH}$, $\mathrm{NADP^+/NADPH}$, ATP/ADP, CoA, acetyl-CoA, and SAM. These additional species introduce long SMILES segments on both the reactant and product sides, causing otherwise simple biochemical transformations to exceed the sequence length required for organic reaction prediction that lead to the insufficeint tokenizations.

To better accommodate long cofactor-containing enzymatic reactions, we initialized a LongT5 encoder-decoder model using weights from the pretrained ReactionT5v2-forward model in addition to the SFT. The LongT5 [18] configuration was matched to ReactionT5 wherever possible, including the vocabulary size, hidden dimension, feed-forward dimension, attention heads, encoder-decoder depth, activation function, and special token identifiers. The original ReactionT5 tokenizer was retained, so the adaptation extended the model architecture for longer sequence handling without changing the tokenization scheme.

Parameters with identical names and tensor shapes were copied directly from ReactionT5 to LongT5. Because LongT5 replaces standard encoder self-attention with local transient-global attention, compatible ReactionT5 self-attention weights were remapped to the corresponding LongT5 local attention modules. LongT5-specific transient-global parameters that were absent from ReactionT5 were left randomly initialized:

$$\theta^{(p)}_{\mathrm{LT5}} \leftarrow \theta^{(p)}_{\mathrm{RT5}} \quad \text{if} \quad \mathrm{shape}(\theta^{(p)}_{\mathrm{LT5}}) = \mathrm{shape}(\theta^{(p)}_{\mathrm{RT5}}).$$

The transferred model was used as the initialization for long-context fine-tuning on metabolic reactions, with increased maximum lengths for both input and target sequences. This preserved the chemistry-aware representations learned by ReactionT5 while allowing the model to process longer reaction strings containing cofactors. This architectural change improves coverage of long enzymatic reaction strings but does not explicitly enforce elemental or stoichiometric balance.

3 Results

Model performance was evaluated using all-product accuracy, any-product accuracy, chemical validity, Tanimoto similarity, and stoichiometric balance score (see Evaluation Details section) for all three different tuning method.

3.1 Metrics

The pretrained baseline generated mostly valid SMILES, but its low product accuracy, low mean

Tanimoto similarity [19], and low stoichiometric balance score show that chemical validity alone was not sufficient for metabolic reaction prediction. This behavior suggests that the pretrained model preserved general reaction syntax from organic chemistry pretraining, but did not capture the substrate-product mappings, cofactor usage, and multi-product structure of biochemical reactions.

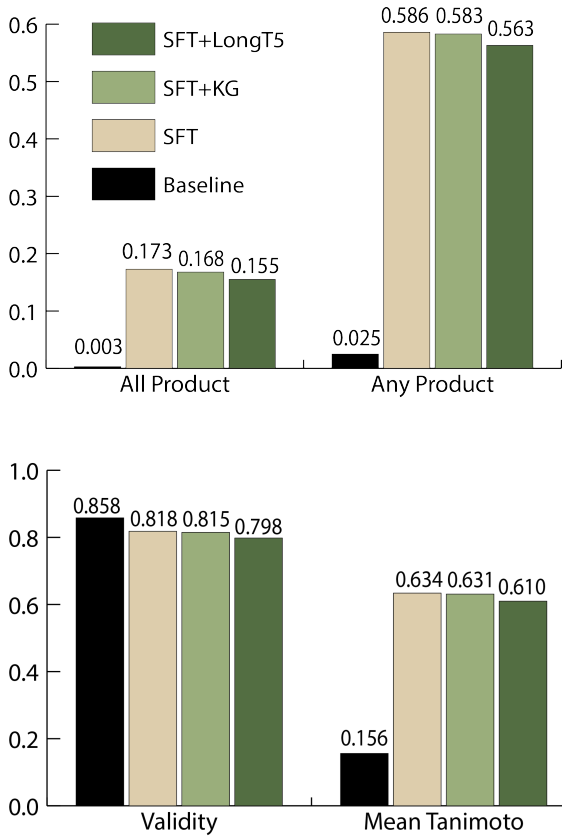


Figure 2: Top-1 reaction prediction performance across the pretrained baseline and adapted models (n=3,388). Each model generated one prediction per reaction. All-product accuracy requires exact product-set recovery without explicit stoichiometric balance enforcement, whereas any-product accuracy requires at least one correctly predicted product.

Among the adapted models, standard SFT produced the strongest overall performance. It achieved the highest product recovery and improved both structural similarity and stoichiometric balance relative to the baseline. In contrast, adding the functional group knowledge graph did not improve top-1 prediction accuracy beyond SFT, although it maintained similar product-level behavior. Similarly, the LongT5 adaptation improved substantially over the baseline but did not outperform standard SFT. Therefore, the results indicate that the main performance gain came from supervised fine-tuning on curated metabolic reactions, while the additional knowledge graph (KG) and LongT5 components did not provide measurable improvement under the current top-1 evaluation setting.

The tuning strategies were motivated by the need to improve not only product recovery but also chemically consistent reaction generation. In metabolic reaction prediction, a correct output should ideally recover the expected products while also satisfying elemental and stoichiometric balance, because an unbalanced reaction may identify a plausible transformation pattern but still be chemically incomplete. Therefore, we separately evaluated the stoichiometric balance score, which measures whether the total elemental composition of the predicted reaction is conserved across the reactant and product sides.

Table 1: Stoichiometric and elementally balance score for the pretrained baseline and adapted models.

Model	**Score**
Baseline	0.033
SFT	0.125
SFT+KG	0.125
SFT+LongT5	0.095

As shown in Table 1, the stoichiometric balance results indicate that implicit learning was insufficient to improve elemental and stoichiometric correctness to the same extent as product-recovery metrics when the tuning datasize is small. Although SFT substantially improved all-product and any-product accuracy, the balance score remained low, increasing only from 0.033 for the baseline to 0.125 for SFT. The SFT+KG model achieved the same balance score as SFT, suggesting that the functional group prior did not impose full-reaction conservation. Similarly, SFT+LongT5 remained below SFT, indicating that longer-context modeling alone was insufficient to ensure balanced reactions. The supervised adaptation improves product prediction, but elemental and stoichiometric correctness require explicit balance-aware constraints, post-generation correction, or atom-mapping-based validation.

3.2 Element-wise Balance Analysis

Element-wise balance was evaluated to identify which atoms contribute most to stoichiometric errors in the generated reactions. This analysis

is important because a reaction can recover the correct product or a structurally similar product while still violating elemental conservation. For each element, the balance accuracy measures the fraction of top-1 predictions in which that element has the same total count on the reactant and product sides. Therefore, higher element-wise balance indicates that the model better preserves that element during reaction generation.

As shown in Figure 3, the adapted models improved element-wise balance relative to the pretrained baseline, but the improvement depended strongly on the element. This analysis is useful because full stoichiometric balance requires every element to be conserved simultaneously, whereas element-wise balance identifies which atoms are responsible for most imbalance errors.

The highest balance accuracies were observed for N, P, and S. These elements often appear in conserved biochemical groups, such as nucleotide cofactors, phosphate groups, and CoA-containing carriers. Their occurrence is more structurally distinctive in SMILES strings, so once the model learns recurring cofactor and carrier patterns, these elements are more likely to be preserved. In contrast, H, O, and C remained more difficult to balance. Hydrogen imbalance was the most prominent source of error because hydrogen is often treated implicitly in biochemical reaction records. Many reactions involve proton transfer, hydride transfer, or pH-dependent protonation states, but these processes may not be represented consistently in SMILES strings or database annotations. For example, redox reactions involving $\mathrm{NAD^+/NADH}$ or $\mathrm{NADP^+/NADPH}$ require precise accounting of hydride and proton equivalents. If the model predicts the correct substrate-to-product conversion but omits $\mathrm{H^+}$, uses the wrong redox cofactor form, or predicts a different protonation state, the reaction becomes hydrogen-imbalanced even when the main product is chemically reasonable.

Oxygen balance is also difficult because oxygen-containing species frequently enter or leave biochemical reactions through water, phosphate, hydroxyl, carbonyl, and carboxylate groups. Hydrolysis, condensation, phosphorylation, and dephosphorylation reactions are especially sensitive to whether $\mathrm{H_2O}$, $\mathrm{P_i}$, ATP/ADP, or oxygenated functional groups are represented correctly. Therefore, a missing water molecule, an incomplete phosphate-containing cofactor, or an incorrect hydroxyl/carboxylate state can produce oxygen imbalance even when the predicted transformation type is plausible.

Carbon imbalance reflects a different type of error. Unlike hydrogen and oxygen, carbon is less affected by implicit protonation or solvent conventions and is more directly tied to whether the model predicts the correct carbon-containing products. Carbon imbalance therefore often indicates missing products, extra carbon-containing fragments, incomplete cofactor or carrier structures, or an incorrect main metabolite. Thus, low carbon balance is more closely associated with failures in product-set recovery, whereas hydrogen and oxygen imbalance can also arise from subtle biochemical bookkeeping errors.

The element-wise results show that supervised adaptation learned some recurring biochemical conservation patterns, especially for cofactor-associated elements, but did not learn a general conservation law. The similar behavior of SFT and SFT+KG further suggests that functional group-level priors can capture reaction-type patterns but do not enforce atom-level conservation.

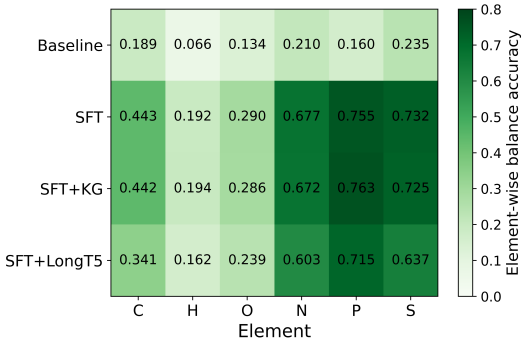


Figure 3: Element-wise balance accuracy across top-1 predictions. Each cell reports the fraction of predictions in which the corresponding element had the same total count on the reactant and product sides.

4 Discussion

The comparison among the three adaptation strategies clarifies both the promise and the limitations of adapting organic reaction language models to biochemical reaction prediction. Standard supervised fine-tuning was the most effective component. Starting from ReactionT5, a model pretrained on organic reaction prediction, SFT substantially improved product recovery after exposure to curated MetaNetX reactions. This is notable because the biochemical tuning dataset was

relatively small compared with typical organic reaction pretraining corpora. Although the processed dataset contained approximately $35{,}000$ training entries, many reactions were duplicated as forward and reverse directions, leaving roughly $19{,}000$ unique MetaNetX transformations for effective fine-tuning. Therefore, the observed improvement suggests that reaction-specific language models already encode useful chemical grammar and can be efficiently transferred to metabolic reaction prediction with limited domain-specific data.

At the same time, the results show that additional model components did not improve performance beyond standard SFT under the current setting. The functional group knowledge graph was introduced to encode chemically interpretable transformation priors, including functional group roles and reaction-type patterns such as oxidation, hydrolysis, condensation, and group transfer. However, using this functional group-level information to guide reaction transformation did not lead to a measurable improvement in top-1 product accuracy or stoichiometric balance. This suggests that the KG captures local chemical function and transformation plausibility, but does not directly solve the full reactant-to-product mapping problem. In practice, the KG may be more useful for interpreting generated reactions, validating reaction classes, filtering chemically unreasonable candidates, or reranking multiple predictions rather than improving single-shot product generation through the current supervised objective.

The LongT5 adaptation was motivated by the length mismatch between organic and enzymatic reaction strings. Enzymatic reactions often include large cofactors and carrier molecules, such as $\mathrm{NAD^+/NADH}$, $\mathrm{NADP^+/NADPH}$, ATP/ADP, CoA, acetyl-CoA, and SAM, which substantially increase SMILES sequence length. LongT5 was therefore used to better accommodate long cofactor-containing inputs. However, the LongT5 variant did not improve over standard SFT and slightly reduced several metrics. A likely reason is that the LongT5 architecture contains parameters absent from ReactionT5, especially transient-global attention components, which remained randomly initialized after weight transfer. Thus, although LongT5 provides longer-context capacity, the benefit of this architecture may require substantially more biochemical training data or more careful pretraining. In the current limited-data setting, the architectural mismatch may have disrupted some chemistry-aware representations inherited from ReactionT5, leading to weaker predictions.

These findings indicate that the primary performance gain came from supervised exposure to metabolic reactions rather than from additional functional group priors or longer-context modeling. This conclusion is also relevant to current efforts that use general-purpose large language models, such as Gemma-style models, for biochemical reasoning. General LLMs provide broad contextual knowledge, but they are not inherently optimized for SMILES translation, reaction grammar, or atom-level chemical transformations. In contrast, reaction-specific models such as ReactionT5 are narrower and less general, but they directly encode molecular string syntax and reactant-product transformation patterns. Our results suggest that these chemistry-specialized models remain strong and efficient baselines for biochemical reaction prediction.

However, SFT alone is not sufficient for chemically complete metabolic reaction generation. The adapted models improved product accuracy and structural similarity, but stoichiometric balance remained low. This shows that token-level likelihood training can learn frequent product patterns without learning a strict conservation law. Reliable biochemical prediction will therefore require coupling reaction-model fine-tuning with explicit biochemical constraints, including atom-mapping validation, balance-aware decoding, post-generation stoichiometric correction, cofactor-aware reranking, and constraint-based filtering. Such additions are necessary to move from plausible product-string prediction toward fully balanced and biologically meaningful metabolic reaction generation.

References

[1] Sébastien Moretti, Van Du T. Tran, Florence Mehl, Mark Ibberson, and Marco Pagni. MetaNetX/MNXref: Unified namespace for metabolites and biochemical reactions in the context of metabolic models. *Nucleic Acids Research*, 49 (D1):D570–D574, 2021. doi: 10.1093/nar/gkaa992.

[2] Akhil Kumar, Lin Wang, Chiam Yu Ng, and Costas D. Maranas. Pathway design using de novo steps through uncharted biochemical spaces. *Nature Communications*, 9:184, 2018. doi: 10.1038/s41467-017-02362-x.

[3] Vikas Upadhyay, Mohit Anand, and Costas D. Maranas. novoStoic2.0: An integrated framework for pathway synthesis, thermodynamic evaluation, and enzyme selection. *PLOS Computational Biology*, 21(8):e1012516, 2025. doi: 10.1371/journal.pcbi.1012516.

[4] Vassily Hatzimanikatis, Chunhui Li, Justin A. Ionita, Christopher S. Henry, Matthew D. Jankowski, and Linda J. Broadbelt. Exploring the diversity of complex metabolic networks. *Bioinformatics*, 21 (8):1603–1609, 2005. doi: 10.1093/bioinformatics/bti213.

[5] Baudoin Delépine, Thomas Duigou, Pablo Carbonell, and Jean-Loup Faulon. RetroPath2.0: A retrosynthesis workflow for metabolic engineers. *Metabolic Engineering*, 45:158–170, 2018. doi: 10.1016/j.ymben.2017.12.002.

[6] Thomas Duigou, Mathilde du Lac, Pablo Carbonell, and Jean-Loup Faulon. RetroRules: A database of reaction rules for engineering biology. *Nucleic Acids Research*, 47(D1):D1229–D1235, 2019. doi: 10.1093/nar/gky940.

[7] Thomas Duigou, Philippe Meyer, and Jean-Loup Faulon. RetroRules 2026: An expanded database combining biochemical and organic reaction templates for pathway discovery. *Nucleic Acids Research*, 54(D1):D1799–D1806, 2026. doi: 10.1093/nar/gkaf1261.

[8] David Weininger. SMILES, a chemical language and information system. 1. introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences*, 28(1):31–36, 1988. doi: 10.1021/ci00057a005.

[9] Philippe Schwaller, Teodoro Laino, Théophile Gaudin, Peter Bolgar, Christopher A. Hunter, Costas Bekas, and Alpha A. Lee. Molecular transformer: A model for uncertainty-calibrated chemical reaction prediction. *ACS Central Science*, 5(9):1572–1583, 2019. doi: 10.1021/acscentsci.9b00576.

[10] Ross Irwin, Spyridon Dimitriadis, Jiazhen He, and Esben Jannik Bjerrum. Chemformer: A pre-trained transformer for computational chemistry. *Machine Learning: Science and Technology*, 3(1):015022, 2022. doi: 10.1088/2632-2153/ac3ffb.

[11] Jieyu Lu and Yingkai Zhang. Unified deep learning model for multitask reaction predictions with explanation. *Journal of Chemical Information and Modeling*, 62(6):1376–1387, 2022. doi: 10.1021/acs.jcim.1c01467.

[12] Tatsuya Sagawa and Ryosuke Kojima. ReactionT5: A pre-trained transformer model for accurate reaction prediction with limited data. *Journal of Cheminformatics*, 17:94, 2025. doi: 10.1186/s13321-025-01075-4.

[13] Yafeng Deng, Xinda Zhao, Hanyu Sun, Yu Chen, Xiaorui Wang, Xi Xue, Liangning Li, Jianfei Song, Chang-Yu Hsieh, Tingjun Hou, Xiandao Pan, Taghrid Saad Alomar, Xiangyang Ji, and Xiaojian Wang. RSGPT: A generative transformer model for retrosynthesis planning pre-trained on ten billion datapoints. *Nature Communications*, 16:7012, 2025. doi: 10.1038/s41467-025-62308-6.

[14] Steven M. Kearnes, Michael R. Maser, Michael Wleklinski, Anton Kast, Abigail G. Doyle, Spencer D. Dreher, Joel M. Hawkins, Klavs F. Jensen, and Connor W. Coley. The open reaction database. *Journal of the American Chemical Society*, 143(45):18820–18826, 2021. doi: 10.1021/jacs.1c09820.

[15] Chonghuan Zhang, Adarsh Arun, and Alexei A. Lapkin. Completing and balancing database excerpted chemical reactions with a hybrid mechanistic-machine learning approach. *ACS Omega*, 9(16):18385–18399, 2024. doi: 10.1021/acsomega.4c00262. URL https://doi.org/10.1021/acsomega.4c00262.

[16] Thao Nguyen, Kuan-Hao Huang, Ge Liu, Martin D. Burke, Ying Diao, and Heng Ji. FARM: Enhancing molecular representations with functional group awareness. *arXiv preprint arXiv:2410.02082*, 2026.

[17] Greg Landrum et al. RDKit: Open-source cheminformatics software. https://www.rdkit.org, 2024.

[18] Mandy Guo, Joshua Ainslie, David Uthus, Santiago Ontañón, Jianmo Ni, Yun-Hsuan Sung, and Yinfei Yang. LongT5: Efficient text-to-text transformer for long sequences. *arXiv preprint arXiv:2112.07916*, 2021.

[19] Dávid Bajusz, Anita Rácz, and Károly Héberger. Why is tanimoto index an appropriate choice for fingerprint-based similarity calculations? *Journal of Cheminformatics*, 7:20, 2015. doi: 10.1186/s13321-015-0069-3.

5 Additional Evaluation Details

We evaluated model performance using five complementary metrics: all-product accuracy, any-product accuracy, chemical validity, Tanimoto similarity, and stoichiometric balance score.

All-product accuracy was defined as

$$\mathrm{Acc}_{\mathrm{all}} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left[\hat{\mathcal{P}}_i = \mathcal{P}_i\right],$$

where $\hat{\mathcal{P}}_i$ is the predicted product set, $\mathcal{P}_i$ is the reference product set, and $\mathbb{I}[\cdot]$ is the indicator function. A prediction was counted as correct only when the complete predicted product set matched the reference product set after preprocessing. No

elemental or stoichiometric balance constraint was applied in this metric.

Any-product accuracy was defined as

$$\mathrm{Acc}_{\mathrm{any}} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left[\hat{\mathcal{P}}_i \cap \mathcal{P}_i \neq \varnothing\right].$$

This metric counts a prediction as correct if at least one predicted product matches any product in the reference product set.

Chemical validity was defined as

$$\mathrm{Validity} = \frac{1}{M}\sum_{j=1}^{M}\mathbb{I}\left[\mathrm{RDKit}(\hat{p}_j) \neq \varnothing\right],$$

where $\hat{p}_j$ denotes a generated product molecule. This metric measures the fraction of generated products that can be parsed as chemically valid molecular structures.

Tanimoto similarity was computed as

$$\mathrm{Tan}(\hat{p}, p) = \frac{\mathbf{b}(\hat{p}) \cdot \mathbf{b}(p)}{\|\mathbf{b}(\hat{p})\|_1 + \|\mathbf{b}(p)\|_1 - \mathbf{b}(\hat{p}) \cdot \mathbf{b}(p)},$$

where $\mathbf{b}(\hat{p})$ and $\mathbf{b}(p)$ are molecular fingerprint vectors for the predicted and reference products, respectively.

The stoichiometric balance score was defined by comparing elemental composition vectors on the reactant and product sides:

$$S_{\mathrm{bal}} = \mathbb{I}\left[\mathbf{e}_{\mathrm{reactants}} = \mathbf{e}_{\mathrm{products}}\right],$$

where $\mathbf{e}_{\mathrm{reactants}}$ and $\mathbf{e}_{\mathrm{products}}$ denote the total elemental composition vectors of the reactant and product sides, respectively. A reaction receives a score of 1 only when the predicted reaction is elementally balanced and 0 otherwise.

5.1 Training loss

The training curves in Figure 4 show that all three adapted models optimized stably during fine-tuning. Training loss decreased consistently for SFT, SFT+KG, and SFT+LongT5, indicating that each model was able to fit the curated metabolic reaction data. The evaluation loss dropped rapidly during the early training phase and then gradually increased after reaching a minimum, suggesting that extended training improved fitting to the training set but did not necessarily improve generalization. This trend is also consistent with the evaluation accuracy curves, where SFT and SFT+LongT5 approached a plateau near the later

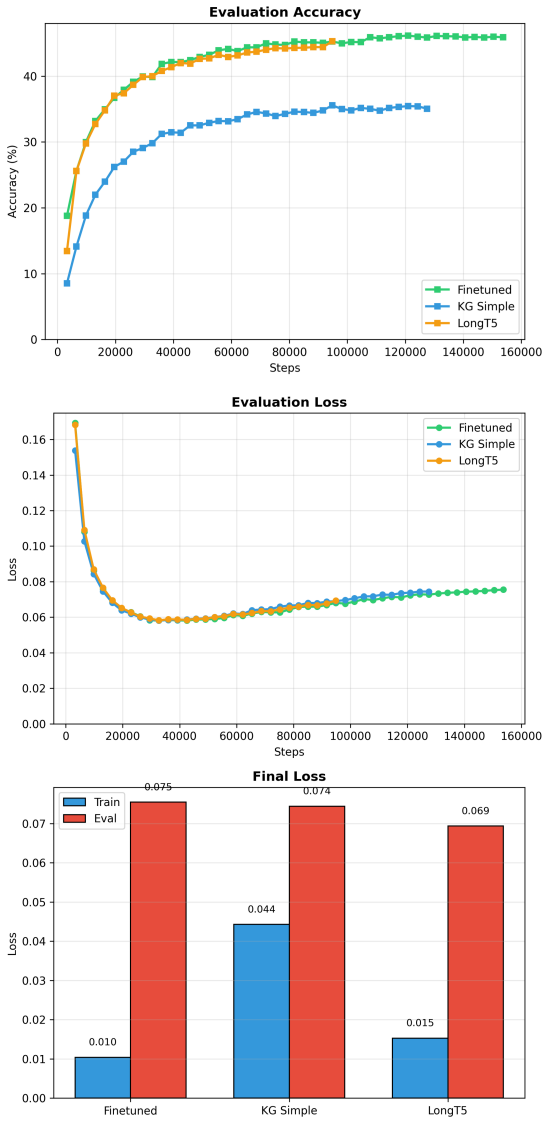


Figure 4: Training dynamics for SFT, SFT+KG, and SFT+LongT5.

training steps, while SFT+KG plateaued at a lower accuracy.

Among the three methods, SFT showed the strongest validation behavior, with the highest evaluation accuracy and the lowest final training loss. SFT+LongT5 followed a similar trend but did not exceed standard SFT, suggesting that longer-context capacity alone did not provide an immediate advantage under the current data and training setup. SFT+KG showed slower improvement and a lower final evaluation accuracy, indicating that the additional knowledge graph-guided component did not improve sequence prediction performance in this implementation. The training curves support the main result where supervised fine-tuning was the dominant source of performance improvement.